

EFTRA-BROEN V, Sweden

WMO: 026030

Lat: 56.8625N

Lon: 12.6628E

Elev: 33

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.1	13.3	5.2	7.2	9.1	9.6	9.0	15.2	23.2	41.3	20.8	40.7	2.2	140	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	80.5	67.0	77.5	65.6	74.2	63.8	68.8	77.1	66.9	74.8	65.2	72.1	6.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	95.3	72.4	63.9	89.2	70.1	62.4	84.4	68.7	33.0	77.7	31.4	74.8	30.2	72.2	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	16.3	14.5	2.4	85.2	6.9	2.7	-2.5	87.1	-6.6	88.7	-10.4	90.2	-15.4	92.2
			2.9	72.9	6.9	3.8	-2.1	75.7	-6.1	77.9	-10.0	80.0	-15.0	82.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.5	30.7	31.7	36.0	44.6	52.2	58.0	63.1	62.6	55.8	47.9	40.8	34.2	(6)
(7)		DBStd	13.04	8.23	6.77	6.73	6.07	5.79	5.02	4.52	4.42	4.88	6.06	6.87	8.92	(7)
(8)		HDD50	2665	600	513	436	182	37	2	0	0	9	113	282	491	(8)
(9)		HDD65	6813	1065	933	900	611	398	218	93	102	279	531	727	956	(9)
(10)		CDD50	1399	0	0	0	22	106	241	407	390	182	47	4	0	(10)
(11)		CDD65	72	0	0	0	0	2	7	35	27	1	0	0	0	(11)
(12)		CDH74	724	0	0	0	1	30	144	361	178	10	0	0	0	(12)
(13)		CDH80	97	0	0	0	0	1	13	64	19	0	0	0	0	(13)
(14)	Wind	WSAvg	6.8	7.7	7.5	7.0	6.7	6.9	6.0	5.7	6.0	6.7	6.7	7.7	7.6	(14)
(15)	Precipitation	PrecAvg	39.0	3.3	2.5	2.2	2.1	2.5	3.5	3.9	4.4	3.5	4.0	3.6	3.4	(15)
(16)		PrecMax	50.2	7.2	5.6	5.7	4.3	4.6	7.1	8.3	7.6	7.5	9.1	6.1	7.9	(16)
(17)		PrecMin	27.2	0.0	0.3	0.6	0.3	0.5	0.0	0.3	1.0	1.0	1.2	1.7	0.4	(17)
(18)		PrecStd	6.3	1.9	1.3	1.2	1.2	1.0	1.7	2.1	1.7	1.7	2.0	1.2	2.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.5	46.6	59.7	70.0	78.0	81.7	85.1	82.3	75.5	63.0	55.4	49.7	(19)
(20)		MCWB	44.5	43.2	48.5	55.1	64.0	65.8	68.5	68.0	63.7	57.3	53.6	48.2	(20)	
(21)		2%	DB	44.7	43.7	51.2	65.5	72.7	78.1	81.6	78.8	70.2	59.8	53.1	47.8	(21)
(22)		MCWB	43.2	42.1	44.9	52.7	58.6	63.9	67.6	66.6	61.5	56.2	51.2	46.2	(22)	
(23)		5%	DB	42.9	42.1	47.0	60.7	68.6	74.2	78.4	75.3	66.6	57.9	51.3	46.5	(23)
(24)		MCWB	41.7	40.6	42.6	50.1	58.4	62.1	67.0	65.7	60.1	55.0	49.5	45.0	(24)	
(25)		10%	DB	41.2	40.4	44.1	56.8	63.8	69.9	74.5	71.4	64.1	56.3	49.8	45.0	(25)
(26)		MCWB	40.1	38.9	41.0	48.1	55.5	60.2	65.0	64.1	58.6	53.8	48.1	43.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	44.1	48.9	56.6	70.1	67.8	71.9	71.1	64.3	58.6	54.0	48.7	(27)
(28)		MADB	46.4	45.5	57.8	68.1	70.9	79.1	80.3	78.5	72.0	60.6	55.0	49.6	(28)	
(29)		2%	WB	43.3	42.4	45.6	53.9	62.2	65.2	69.2	68.1	62.6	57.2	51.7	46.6	(29)
(30)		MADB	44.1	43.5	50.5	63.4	67.7	75.1	78.5	75.0	68.8	59.2	52.9	47.6	(30)	
(31)		5%	WB	41.8	40.8	43.3	51.3	58.3	63.3	67.5	66.3	61.0	55.7	50.0	45.3	(31)
(32)		MADB	42.7	41.9	46.1	59.1	66.0	71.5	76.4	72.8	65.4	57.5	51.3	46.3	(32)	
(33)		10%	WB	40.2	39.1	41.6	48.8	55.7	61.3	65.9	64.7	59.5	54.2	48.3	43.5	(33)
(34)		MADB	41.1	40.4	43.5	55.2	62.3	68.6	73.3	70.4	63.2	56.1	49.6	44.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.4	9.0	12.7	17.1	16.3	17.0	16.7	14.9	14.3	11.7	8.1	7.6	(35)	
(36)		5% DB	MADB	6.3	7.7	17.3	25.1	24.9	27.1	25.6	21.6	19.1	11.9	7.0	6.0	(36)
(37)		MCWBR	6.4	6.3	11.8	14.6	14.7	15.0	14.0	11.4	11.5	8.9	6.4	6.0	(37)	
(38)		5% WB	MADB	6.1	6.6	14.5	22.3	22.0	22.6	22.1	18.2	16.4	10.0	6.5	5.8	(38)
(39)		MCWBR	6.4	5.8	10.5	13.8	14.4	13.5	13.1	10.9	10.6	8.2	6.1	6.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.308	0.333	0.352	0.388	0.385	0.375	0.399	0.399	0.367	0.352	0.330	0.297	(40)	
(41)		taud	2.267	2.305	2.332	2.291	2.370	2.421	2.373	2.376	2.448	2.396	2.314	2.234	(41)	
(42)		Ebn at Noon	180	226	254	262	271	275	265	256	248	217	170	149	(42)	
(43)		Edh at Noon	16	24	30	36	35	34	35	33	26	22	16	13	(43)	
(44)	All-Sky Solar Radiation	RadAvg	126	316	751	1243	1626	1724	1592	1272	891	422	158	87	(44)	
(45)		RadStd	25	50	76	170	155	140	198	116	106	66	29	10	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (65 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	-227	-320	N/A	N/A	(47)

Nomenclature: See separate page